

Brainstorming the architecture of asset management

- Goal of document
 - Serve as the project technical terminology
 - Serve as an internal baseline for brainstorming and communications in between team members
 - so, only to the extent that the team members get in the same page of discussion
 - neither to the extent of publish quality nor educational quality
- Scope of document
 - Requirements, and architectural design deduced directly from the requirements
 - Recommended design decisions and their explanation
 - Major algorithms to debate on

1. Specifying system requirements

- Major concepts and requirements: **Deposit, stake, compound, withdraw**
 - A user is a wallet, technically. If a user has multiple wallets, the user is the wallet that is currently involved with Mozaic by the user's current action.
 - A users **deposits** their assets from their wallet to Mozaic vault wallet, in exchange with LP tokens.
 - Mozaic **stakes** the deposited assets from vault wallet(s) to staking pools. Staking is on its behalf of Mozaic and not of individual users.
 - Mozaic **stakes** rewards coming from its staking on its behalf. Or, Mozaic **compounds** rewards. Note: With compounding, the total assets become a mixture of deposited and rewarded assets.
 - *Mozaic stakes on a collective basis.* All deposited assets are staked and performs collectively. You cannot track which user's deposit is staked on which staking pool and performs how distinctively.
 - *Mozaic stakes on a global basis.* No matter what tokens on what chain a user deposits, Mozaic will convert freely the deposits to listed acceptable token on any listed chain, to maximize profit.
 - **Staking stock** is the sum of total currently staked assets and total currently pending rewards on staking pools.
 - An amount of LP token held in a user's wallet represents the user's proportional share of **Staking stock**.
 - A user's **Stock share** is part of **Staking stock** covered by the LP tokens held in the user's wallet.
 - A user **withdraws** assets from their **Stock share**, on a whole or partial basis, in any listed token they prefer.

Mozaic system has aggregational and DAO use cases.

Aggregational use cases are shown in the following figure:

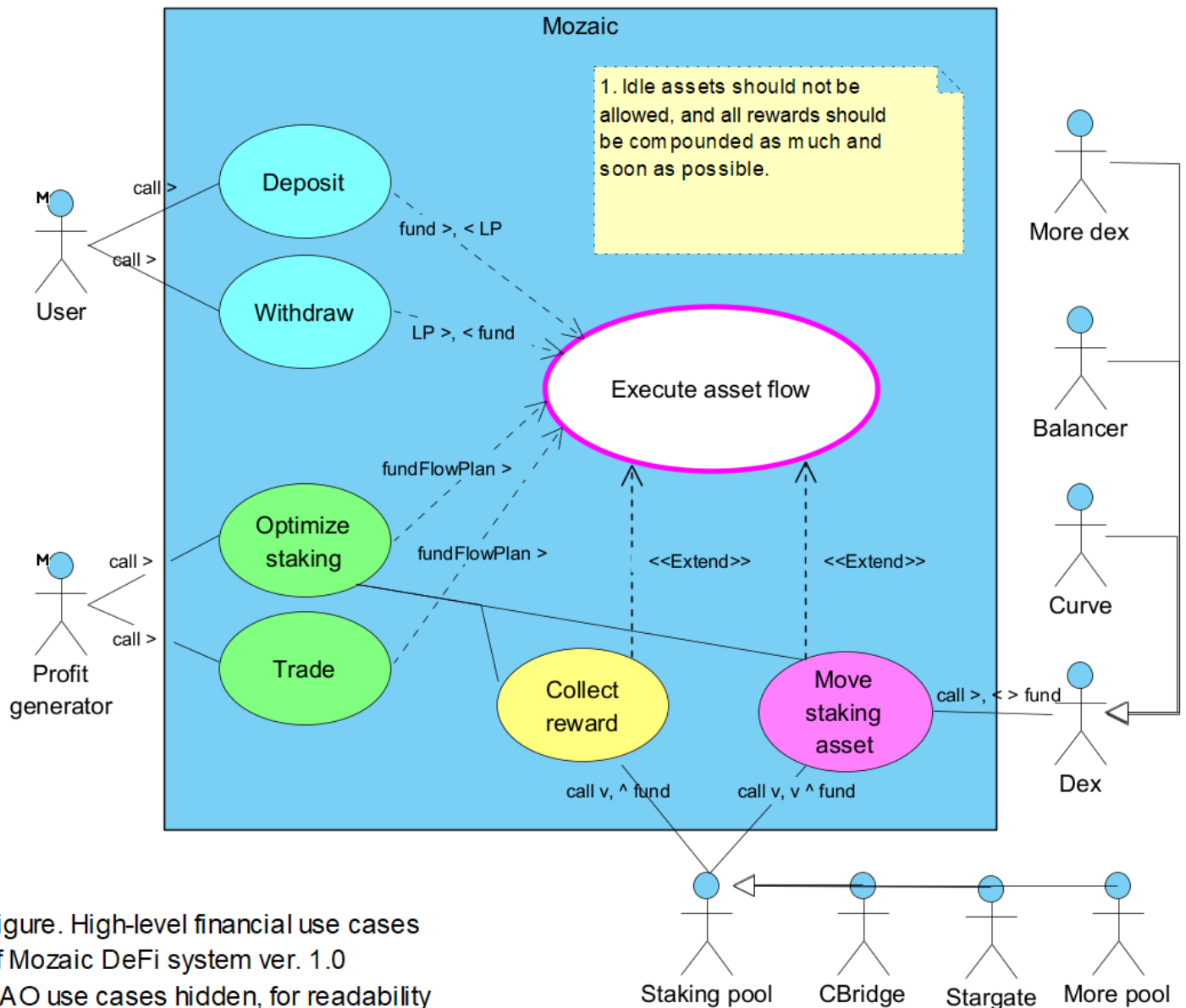


Figure. High-level financial use cases of Mozaic DeFi system ver. 1.0
DAO use cases hidden, for readability

The use cases and external actors are identified as below:

- **Compound:** This hidden use case compounds rewards. Idle assets should not be allowed, and all rewards should be compounded as much and soon as possible.
- **Execute asset flow:** This use case checks, carry out, and keeps track of asset moves. The assets managed by the system can only be moved by this use case transparently.
- **Deposit:** This use case deposits the user's assets in the system. **User** invokes this use case in the hope that Mozaic system will **Optimize staking** of the deposited assets for them. **User** should be able to deposit any token that is listed on Mozaic on any chain where Mozaic is deployed. This use case includes **Execute asset flow**.
- **Withdraw:** This use case withdraws from the user's rewards and, if needed, deposited assets. **User** should be able to withdraw any token that is listed on Mozaic on any chain where Mozaic is deployed, no matter they deposited what token on which chain. (The deposited assets are increased by perpetual compounding of rewards.) **User** invokes this use case. This use case includes **Execute asset flow**.
- **Optimize staking:** This use case upgrades the staking of deposited assets to maximize rewards. **Profit generator**, a role of the system, invokes this use case *either on a regular basis or at randomly picked times*. This use case includes **Execute asset flow** and **Compound**. *By providing **Execute asset flow** with **assetFlowPlan**, this use case effectively prevents it from being involved with finding optimal staking portfolio.*

- **Trade:** This use case swaps idle assets to get profit by using price changes. It calls **Dex**. By providing **Execute asset flow** with **AssetFlowPlan**, this use case effectively prevents it from being involved with finding optimal trading orders.
- **Collect reward:** This use case collects rewards from Staking pools. Use case Execute asset flow, when it is working under **Optimize staking**, is extended by this use case. This use case calls **Staking pool**.
- **Move staking asset:** This use case move assets to/between/from, **Staking pools**. **Execute asset flow**, when it is working under Optimize staking, is extended by this use case. This use case calls **Staking pool**.
- **Dex:** This actor is a smart contract that swaps between assets. Examples are pairs on Curve and Balancer DeFis.
- **Staking pool:** This actor is a smart contract that allocates reward to assets that are staked in it. Examples are farming pools on CBridge and Stargate DeFis.

Asset management of the system has the following State Machine:

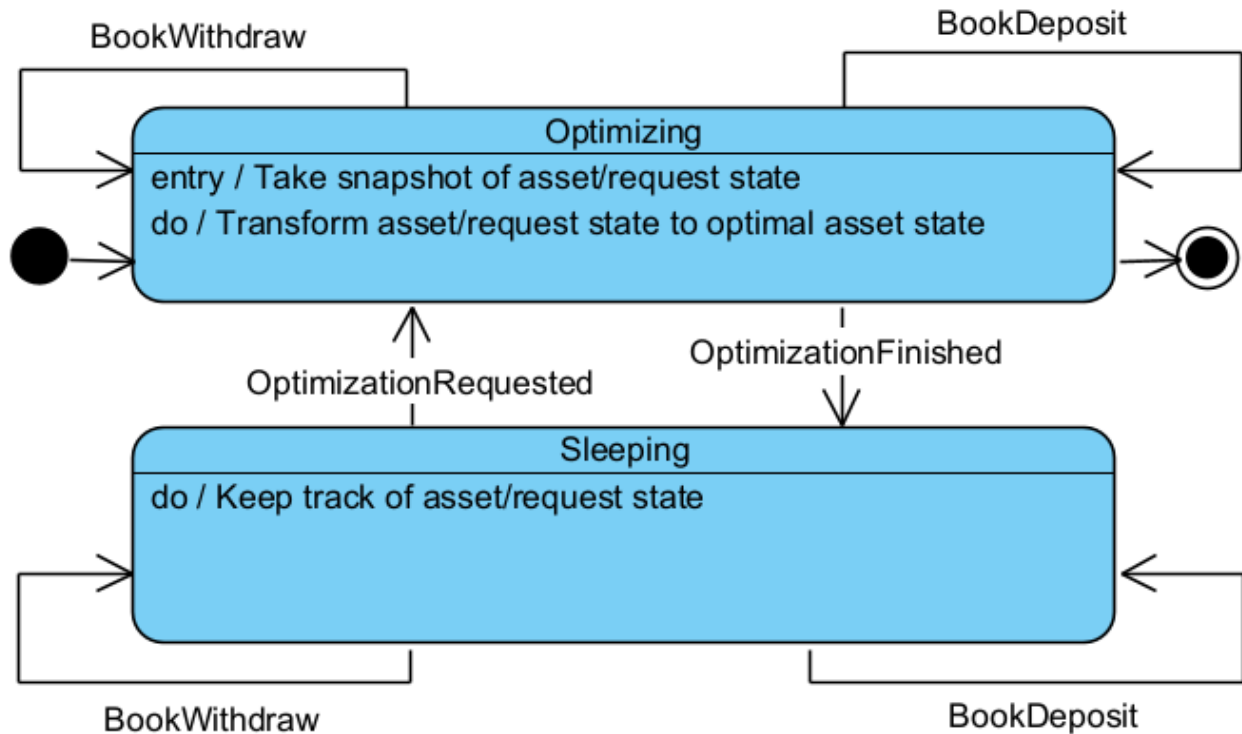


Figure. State Machine of Mozaic asset management

2. Identifying vaults through its surrounding modules

As the first implementation step, we identify the module that executes use case ****Execute asset flow****, as it seems to act as the controller and be one of unique features of the system. The design decisions are as illustrated in the following figure:

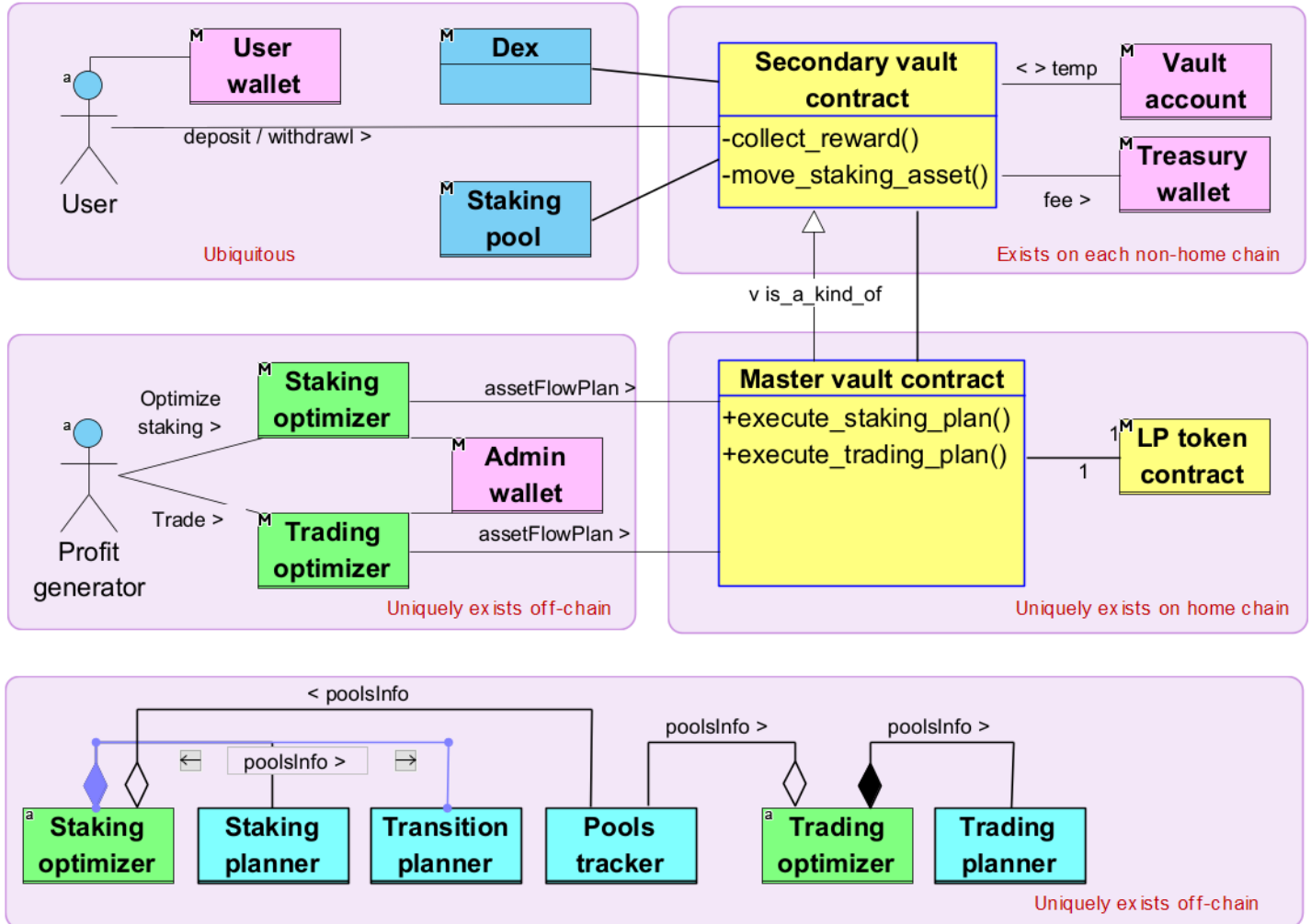


Figure. High-level functional modules of Mozaic DeFi ver. 1.0

Goal: Minimize the functionality of vaults.

Functional modules are described below:

- **Secondary valut contract:** This module is a smart contract and deployed on each chain except the home chain. It participates in the cooperation with its peer **Secondary vault contracts** to implement **Execute asset flow**, This module has at least the following private operations that work locally on its chain: `collect_reward(...)` and `move_staking_asset(...)`
- **Master vault contract:** This module is a smart contract, has global operations, in addition to the local operations inherited from **Secondary vault contract**, and is deployed on one and only one of the chains, called **Home chain**, where it takes the role of **Secondary vault contract**, as well as the the unique role of the master vault operating **LP token contract**.
- **LP token contract:** This module manages the LP token balances of **Users**, which represents the their proportional share of the total assets of the system. Their share comprises of their deposited assets plus automatic compounding. *It is a deliberate design decision that the LP token only exists on **Home chain**.*

- **User wallet:** This is a blockchain wallet and identifies a **User**. **User's** actions; like deposit, withdraw, and harvest; are authenticated/authorized with this wallet.
- **Vault account:** It is the blockchain account of, and controlled by, **Secondary vault contract** and used to store temporary assets, like funds pending staking.
- **Treasury wallet:** This is a blockchain wallet, and a place to store and retrieve system revenues, like fees. It will be better if it is not owned by a human, but be the account of a smart contract that only obeys vault contracts, for better decentralization. It is deployed on all chains.
- **Staking optimizer:** This is an off-chain module that can invoke **Master vault contract**. This module is globally unique, calculates optimal **assetFlowPlans**, and lets the master vault to execute the plans (in cooperation with secondary vaults). *It is an important design decision that the assetFlowPlan is calculated off-chain, thus leading to transparency and security debates, for the sake of gas- and time- savings:*
 - Transparency debate: **Users** will not be able to track why the system chose particular **assetFlowPlans** technically.
 - Security debate: If the calculation of **assetFlowPlan** is hacked or compromised, then the system will make a less-optimal staking maneuver.
 - Justification: Only the second of the following concerns becomes less transparent, leading to both un-assured best profitability and assured huge gas- and time- savings.
 - how much of what assets from which pool to which pool, is the move about
 - whether all the asset moves are securely and/or reasonably/optimally chosen
 - whether all the asset moves are securely executed and logged
 - whether the move logs are readily available to check later
 - whether the execution of assetFlowPlan is integrated
- **Trading optimizer:** This off-chain module is similar to **Staking optimizer**, except it relates to trading.
- **Admin wallet:** This wallet is used to invoke **"Master vault contract"**, in privilege, on behalf of the administrator.
- **Staking planner:** An integral component of **Staking optimizer**, this module predicts the next most profitable **staking_portfolio**, based on **poolsInfo** provided by **Pools tracker**. Running this module on-chain would enhance transparency, but would at the same time incur huge gas fee and effectively disable the system.
- **Transition planner:** An integral component of **Staking optimizer**, this module predicts the most efficient **assetFlowPlan**, which is the best procedure of asset move that implements the transitioning to a given **staking_portfolio**, based on the current **poolsInfo**.
- **Trading optimizer:** This is similar to **Staking optimizer**, except that it relates trading.
- **Trading planner:** This is similar to **Staking planner**, except that it relates trading.
- **Pools tracker:** A shared module between **Staking optimizer** and **Trading optimizer**, this module retrieves and tracks all relevant information from chains, like Reward Release Speed, and total Staked LP of each pool. Running this module on-chain would enhance transparency, but would at the same time incur huge gas fee and effectively disable the system.

3. Exploring vaults

We have identified vaults through their surrounding modules interacting with them.

The external actors in the following use case diagram, together with their interactions with vaults are already described above.

We can now explore the use cases of vault.

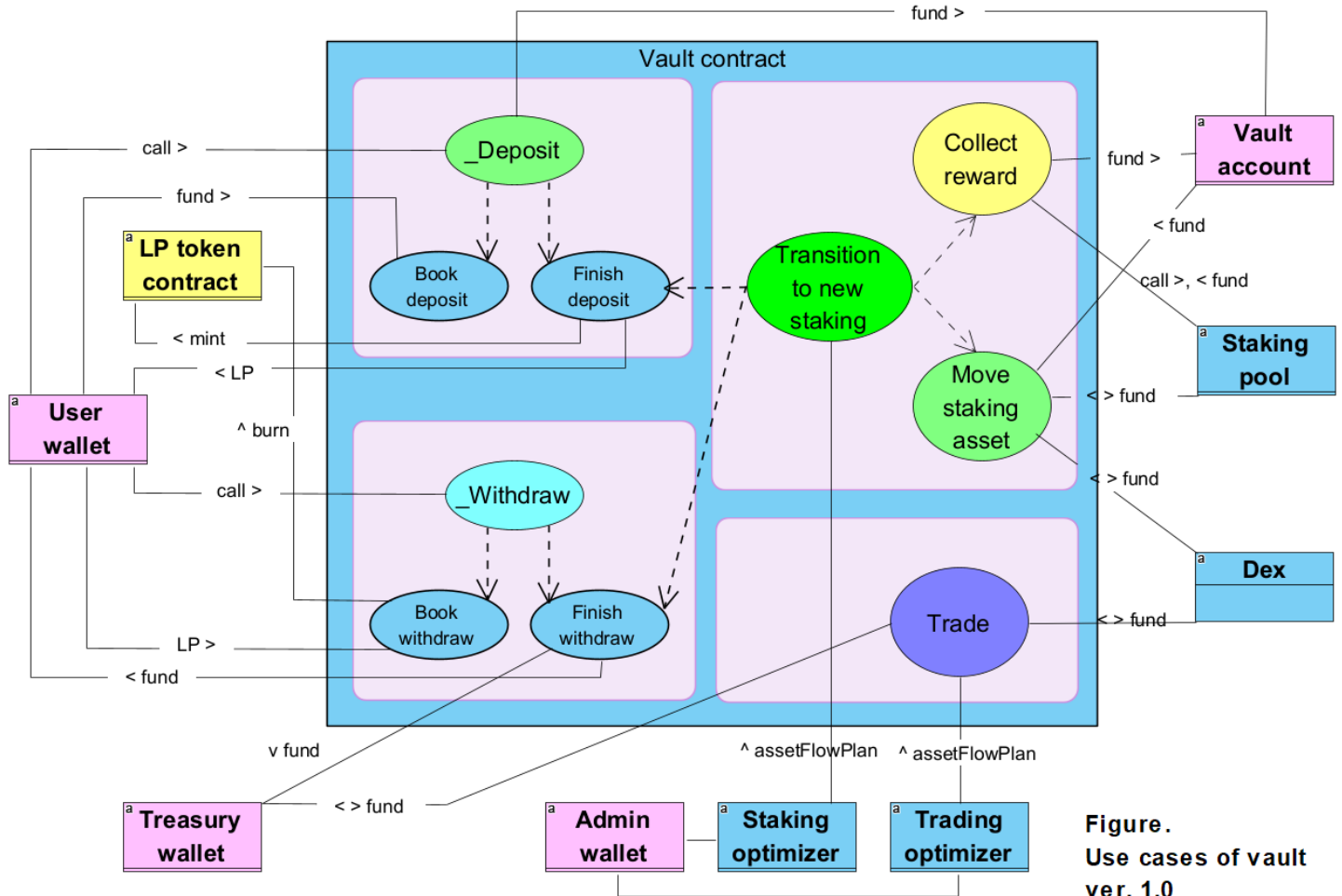


Figure. Use cases of vault ver. 1.0

- **_Deposit**: This is what happens at the level of vault contracts when **Deposit** use case is invoked at the system level. Invoked by **User wallet** with fund amount to deposit, this use case coordinates the following two use cases.
- **Book deposit**: This use case
 - collects the fund from **User wallet**,
 - books the deposit request with the system,
 - and pauses the session of "_Deposit".
- **Finish deposit**: This use case
 - resume the session of "_Deposit",
 - retrieves the booked deposit request,
 - mints LP tokens to cover the new fund,
 - returns the LP tokens to **User wallet**,
 - and helps **Transition to new staking** stake the fund.
- **_Withdraw**: This is what happens at the level of vault contracts when **Withdraw** use case is invoked at the system level. Invoked by **User wallet** with LP tokens returned, this used case coordinates the following two use cases.
- **Book withdraw**: This use case

- collects the returned LP tokens,
- burns the collected LP tokens,
- books the withdrawal request with the system,
- and pauses the session of "_Deposit".
- **Finish deposit:** This use case
 - resume the session of "_Deposit",
 - retrieves the books withdrawal request,
 - subtract fund, as much as covered by the returned LP tokens, from the total system assets, and
 - returns the fund to **User wallet**
- **Transition to new staking:** This use case transitions to a new asset state by executing **assetFlowPlan** provided by **Staking optimizer**. (This is the most challenging part of vault implementation.) It does *collectively*, by using **Move staking asset**,
 - **Collect reward**,
 - Collect staked assets, to cover the fund to **Finish withdraw**,
 - **Finish deposit**,
 - **Finish withdraw**,
 - execute remaining part of **assetFlowPlan**
- **Trade:** This use case executes **AssetFlowPlan** provided by **Trading optimizer**.

4. Algorithm of Staking planner

Note. All errors, like numerical processing rounding and price slippage, are ignored at this stage of architectural design.

4.1 Task definition

- Goal: Calculate best staking portfolio, in order to
 - Save vault contracts long calculations of staking optimization, thus to save gas.
 - Keep vault contract insulated from future algorithm upgrades of staking optimization.
- Consideration
 - Input may not idealistically consistent inside itself, because idealistic snapshot of multiple chain state is impossible logically.
 - Output staking portfolio may not be completely/perfectly implemented, because input may have errors and there are unpredictable price slippages sneaked into the calculation.
- Input
 - Deposit requests currently booked
 - Withdrawal requests currently booked
 - Current pending rewards
 - Current poolsInfo
- Process
(See below)
- Output
 - optimal staking portfolio

4.2 Definition

- **Pools state**

$$PS = \{PS_i | i = 1..N\},$$

where PS_i is the i -th pool state:

$$PS_i = (RewardRate_i, RewardToken_i, TotalStake_i, StakingToken_i, MozaicStake_i)$$

Note: We assume reward on PS_i is calculated as:

$$MozaicReward_i = \frac{RewardRate_i}{TotalStake_i} \times MozaicStake_i$$

- **Token vectors**

- **User tokens vector**

$$UserTokens = \{UserToken_i | i = 1..M\}$$

Example: $UserTokens = \{USDT, USDC, ETH, BNB, BTC\}$

- **Reward tokens vector**

$$RewardTokens = \{RewardToken_i | i = 1..N\}$$

- **Staking tokens vector**

$$StakingTokens = \{StakingToken_i | i = 1..N\}$$

- **Tokens vector**

$$Tokens = (UserTokens, RewardTokens, StakingTokens)$$

- **Asset vectors**

- **Vector of booked deposit requests**, at time index t

$$Deposits^t = \{D_i^t | D_i^t : \text{deposit amount, at time } t, \text{denominated by } UserToken_i, i = 1..M\}$$

Example: $Deposits^t$ of {1, 2, 3} means {1 USDT, 2 USDC, 3 ETH}, assuming $UserTokens = \{USDT, USDC, ETH\}$

- **Vector of booked withdrawals requests**, at time index t

$$Withdrawals^t = \{W_i^t | D_i^t : \text{withdrawal amount, at time } t, \text{denominated by } UserToken_i, i = 1..M\}$$

- **Vector of collected rewards**, at time index t

$$Rewards^t = \{R_i^t | D_i^t : \text{reward amount, at time } t, \text{denominated by } RewardToken_i, i = 1..M\}$$

- **Vector of staked assets**, at time index t

$$Stakes^t = \{S_i^t | D_i^t : \text{stake amount, at time } t, \text{denominated by } StakingToken_i, i = 1..M\}$$

- **Transformations**

- **Transformation $USDT^{+1}$**

$USDT^{+1}$ transforms a Tokens-denominated asset vector to USDT-denominated vector, with market exchange rates.

Example: $USDT^{+1}$ transforms (1, 2, 3) to (1, 2.02, 1300), assuming $UserTokens = (USDT, USDC, ETH)$ and $USDT/USDC = 1.01$ and $USDT/ETH = 1300$.

- **Transformation $USDT^{-1}$**

$USDT^{-1}$ transforms a USDT-denominated asset vector to Tokens-denominated vector, with market exchange rates.

Example: $USDT^{-1}$ transforms (1, 2.02, 1300) to (1, 2, 3), assuming $UserTokens = (USDT, USDC, ETH)$ and $USDT/USDC = 1.01$ and $USDT/ETH = 1300$.

- **Transformation FOP**

FOP , standing for Find Optimal Portfolio, finds the best USDT-denominated vector of staked assets, for a given total USDT amount.

Example: FOP transforms (12345) to the best staking of 123 USDT over staking pools, and has the format of USDT-denominated $Stakes^t$, like (100, 20, 3) all in USDT, assuming we have total 3 pools.

- **Transformation Sum**

$ElementWiseSum$ sums up all elements of a vector.

- **Mozaic's asset state**, at transition t

- Asset snapshot just before the transition that takes place at time t :

$$MS^t = (Tokens, Deposits^t, -Withdrawals^t, Rewards^t, Stakes^t)$$

$$\text{Or, simply, } MS^t = (T, D^t, -W^t, R^t, S^t)$$

- Asset snapshot just after the transition that takes place at time t :

$$MS^{t+} = (Tokens, 0Deposits, -0Withdrawals, 0Rewards, optimal\ Stakes^t)$$

$$\text{Or, simply, } MS^{t+} = (T, 0D, -0W, 0R, optimal\ S^t)$$

, where 0D, 0D, and 0R are a vector of zero values in their respective vector lengths.

4.3 Formulae

If a state transition takes place at time t , Mozaic's asset state MS^t changes to MS^{t+} as shown in the following diagram:

$$\begin{array}{ccc}
MS^t = (T, D^t, -W^t, R^t, S^t) & \xrightarrow{(Resulting\ transition)} & MS^{t+} = (T, 0D, -0W, 0R, optimal\ S^t) \\
\downarrow USDT^{+1} & & \uparrow USDT^{-1} \\
MS_U^t = (T, USDT^{+1}(D^t), \dots) & \xrightarrow{(Implicit)} & MS_U^{t+} = (T, 0D, -0W, 0R, FOP(Total\ in\ USDT)) \\
\downarrow Sum & & \uparrow zeros, FOP \\
Total\ in\ USDT & \xrightarrow{Identity} & Total\ in\ USDT
\end{array}$$

The algorithm for Staking planner MS^t is a chain of transformations:

$$optimal\ S^t = USDT^{-1} \circ FOP \circ Sum \circ USDT^{+1}(T, D^t, -W^t, R^t, S^t)$$

- $USDT^{+1}$ and $USDT^{-1}$ are obvious, except that we may need systematical methods to find best Dexes and swap paths.
- FOP is solved analytically, demonstrating about 9% of competitive edge over the public.
- ElementWiseSum is trivial.
- D^t and W^t can be retrieved from the booked requests of deposits and withdrawals.
- S^t is found when we "Collect reward" pending rewards.

5. Algorithm of Transition planner

5.1 Mozaic LP token will only be present on the home chain.

- LP token can be deployed on any chain(s). According to the requirements, users can deposit any token on any chain and withdraw any token on any chain. It doesn't matter which chain their LP token belongs to.
- If LP token is deployed on multiple chains, there might be significantly bad user experience and cost overhead:
 - A user's wallet may have multiple versions of LP token, although they equally represent the user's, global, Stock share. This poses inconvenience to withdrawal UI and to users. Users will have to think in terms of "how much of what LP tokens to return", instead of "how much LP tokens to return".
 - Suppose the user chooses multiple LP versions to return. Either off-chain modules will have to ask the user to switch the wallet to each of the multiple chains, in turn, to enable the vault on that chain to collect back the LP token on the chain, in an inconvenient and in-secure way; or the user will have to invoke withdrawal process multiple times each focusing on one of the LP versions; in an inconvenient and secure way. Or, the user will have to swap the LP versions into a single LP version, before invoking withdrawal.
 - When depositing, on the other hand, users will have to choose which LP token version to receive, although they equally represent the user's global Stock share. The vault that is serving a user's deposit, will have to send an LP message to the chain where the user-chosen LP version presents, if the user's deposit token and their chosen LP version are on different chains. (Batch messaging might mitigate this a little.)
- If LP token is deployed on multiple chains, we will have to allocate huge initial liquidity on those LP liquidity pairs, for little convenience for traders when trading is not yet important.
- If there is critics that solo LP version, which is naturally deployed on the home chain, will force Mozaic system to transfer to, or send LZ message to, the, non-home, chain on which the user wants to withdraw a token type, then we can say the message will rarely serve a single user but mostly several users in batch, because the withdrawals happen while transitioning to a new staking when all booked deposit requests are handled.

5.2 Task definition for Transition planner

- Goal: Calculate best transition asset flow plan, in order to
 - Save vault contracts long calculations of staking optimization, thus to save gas.
 - Keep vault contract insulated from future algorithm upgrades of staking optimization.
- Consideration
 - Input may not idealistically consistent inside itself, because idealistic snapshot of multiple chain state is impossible logically.
 - Output may not be completely/perfectly implemented, because input may have errors and there are unpredictable price slippages on Dexes.
- Input (see section 4. Algorithm of Staking planner, for their definitions)
 - $Deposits^t$
 - $Withdrawals^t$
 - $Rewards^t$
 - $Stakes^t$
 - $poolsInfo^t$

- *optimal Stakes*^{t+}, or *optimal S*^{t+}
- Process
 - Find a step-by-step logical procedure of asset flow instructions that implements the following transition:

$$MS^t = (T, D^t, -W^t, R^t, S^t) \xrightarrow{(Resulting\ transition)} MS^{t+} = (T, 0D, -0W, 0R, optimal\ S^t)$$
 - Regularize the procedure by Removing mutually/transitively cancelling asset moves in the procedure
 -
- Output
 - *assetFlowPlan*^t

5.3 Algorithm for Transition planner